

# David Ibarra Luna

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## SUMMARY

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Industrial Chemist with a Master's in Materials Science, specialized in nanomaterials synthesis applied to water quality assurance. Experienced in interdisciplinary research combining materials development, contaminant analysis, and data processing with Python, aimed at solving real-world environmental challenges.

## PROFESSIONAL EXPERIENCE

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### Research stay at institutes of CSIC (Barcelona, Spain)

Sep 2024 - Dec 2024

- Developed carbon nanomaterials for optoelectronic nitrate sensors in water.
- Analyzed and processed spectral data from characterization techniques (electron microscopy and spectrophotometry) using Python, accelerating the interpretation of results.

### Robotics teacher (High School)

Aug 2023 - Jun 2025

- Taught robotics and programming (Python) to high school students using the STEM-based FIRST Lego League methodology, fostering technical and problem-solving skills through hands-on projects.

## PROJECTS

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### DIY Optoelectronic Sensor with ML for Water Quality

Ongoing

Building a Raspberry Pi-based optoelectronic sensor for water quality analysis. Implementing ML models (PLS, SVM) in Python to optimize contaminant level detection from spectral data.

### Master's Work: Data Analysis & Animated Visualization

[Data Repo](#) | [Animation Repo](#)

Managed the full data process for a research article, from processing spectral data with Python (Pandas, NumPy) to creating publication-ready figures (Matplotlib). Developed a fully animated thesis defense with Manim to visually communicate complex scientific results ([view presentation here](#)).

### Spectrum Data Loader - Python Package (PyPI)

[Link to Project](#)

Developed and published a Python library on PyPI to streamline the loading and initial processing of spectroscopy data files.

## EDUCATION

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2023–2025 M.Sc. in Materials Chemistry, Autonomous University of Nuevo Leon, (Grade: 94.44/100)

2019–2022 B.Sc. in Industrial Chemistry, Autonomous University of Nuevo Leon, (Grade: 82.69/100)

## PUBLICATIONS

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Ibarra, David et al. (Apr. 2023). "Synthesis of Graphene Quantum Dots Coupled to Au Nanoparticles: A Facile and Versatile Route Using Different Carbon Sources". In: *C* 9.2, p. 45. DOI: [10.3390/c9020045](#).

## SKILLS

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**Computing skills** Python (Pandas, NumPy, Scikit-learn, Matplotlib, ManimCE), Git, GitHub, L<sup>A</sup>T<sub>E</sub>X, ImageJ, MS Office Suite, Inkscape.

**Scientific Skills** Experimental Design, Material Characterization, Quantitative Analysis.

**Languages** Spanish (Native), English (Advanced), French (Beginner).